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ROLL NO:- 2520030188

SEC NO:- 02

1. First Come First Serve

GNU nano 8.7.1

#include <stdio.h>

```
int main() {
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");
    for(i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    wt[0] = 0;

    for(i = 1; i < n; i++)
        wt[i] = wt[i-1] + bt[i-1];

    for(i = 0; i < n; i++)
        tat[i] = wt[i] + bt[i];

    printf("\nProcess\tBT\tWT\tTAT\n");

    for(i = 0; i < n; i++)
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);

    return 0;
}
```

OUTPUT:-

```
praneeth@PRANEETH: ~  
praneeth@PRANEETH:~$ nano fcfs.c  
praneeth@PRANEETH:~$ gcc fcfs.c -o fcfs  
praneeth@PRANEETH:~$ ./fcfs  
Enter number of processes: 3  
Enter burst time:  
P1: 5  
P2: 9  
P3: 11  
  
Process BT      WT      TAT  
P1      5       0       5  
P2      9       5      14  
P3     11     14     25  
praneeth@PRANEETH:~$ nano fcfs.c  
praneeth@PRANEETH:~$ |
```

2. SJF — Non-Preemptive

GNU nano 8.7.1

#include <stdio.h>

```
int main() {
    int n, i, j, temp;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");
    for(i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    /* Sort by burst time */
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    wt[0] = 0;

    for(i = 1; i < n; i++)
        wt[i] = wt[i-1] + bt[i-1];

    for(i = 0; i < n; i++)
        tat[i] = wt[i] + bt[i];
}
```

```

        tat[i] = wt[i] + bt[i];

    printf("\nProcess\tBT\tWT\tTAT\n");

    for(i = 0; i < n; i++)
        printf("P%d\t%d\t%d\t%d\n",
                i + 1, bt[i], wt[i], tat[i]);

    return 0;
}

```

OUTPUT:-

```

praneeth@PRANEETH:~$ nano sjf.c
praneeth@PRANEETH:~$ gcc sjf.c -o sjf
praneeth@PRANEETH:~$ ./sjf
Enter number of processes: 4
Enter burst time:
P1: 1
P2: 9
P3: 5
P4: 77

Process BT      WT      TAT
P1      1        0        1
P2      5        1        6
P3      9        6       15
P4     77       15       92
praneeth@PRANEETH:~$ |

```

3. Priority Scheduling

GNU nano 8.7.1

#include <stdio.h>

```
int main() {
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time and priority:\n");

    for(i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }

    /* Sort according to priority */
    for(i = 0; i < n - 1; i++) {
        for(j = i + 1; j < n; j++) {
            if(priority[i] > priority[j]) {

                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    /* Waiting time */
    wt[0] = 0;

    for(i = 1; i < n; i++)
```

```
wt[0] = 0;

for(i = 1; i < n; i++)
    wt[i] = wt[i-1] + bt[i-1];

/* Turnaround time */
for(i = 0; i < n; i++)
    tat[i] = wt[i] + bt[i];

printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

for(i = 0; i < n; i++)
    printf("P%d\t%d\t%d\t\t%d\t%d\n",
           i + 1, bt[i], priority[i], wt[i], tat[i]);

return 0;
```

OUTPUT:-

```
praneeth@PRANEETH:~$ gcc priority.c -o priority
praneeth@PRANEETH:~$ ./priority
Enter number of processes: 3
Enter burst time and priority:
P1 Burst Time: 2
P1 Priority: 5
P2 Burst Time: 8
P2 Priority: 3
P3 Burst Time: 9
P3 Priority: 1
```

Process	BT	Priority	WT	TAT
P1	9	1	0	9
P2	8	3	9	17
P3	2	5	17	19

4. ROUND ROBIN

GNU nano 8.7.1

#include <stdio.h>

```
int main() {
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for(i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);

        rem_bt[i] = bt[i];
        wt[i] = 0;
    }

    printf("Enter time quantum: ");
    scanf("%d", &quantum);

    /* Round Robin */
    while(1) {
        done = 1;

        for(i = 0; i < n; i++) {
            if(rem_bt[i] > 0) {
                done = 0;

                if(rem_bt[i] > quantum) {
                    time = time + quantum;
                    rem_bt[i] = rem_bt[i] - quantum;
                }
                else {
                    time = time + rem_bt[i];
```

```

    }
    else {
        time = time + rem_bt[i];
        wt[i] = time - bt[i];
        rem_bt[i] = 0;
    }
}

}

if(done == 1)
    break;
}

/* Turnaround time */
for(i = 0; i < n; i++)
    tat[i] = wt[i] + bt[i];

printf("\nProcess\tBT\tWT\tTAT\n");

for(i = 0; i < n; i++)
    printf("P%d\t%d\t%d\t%d\n",
        i + 1, bt[i], wt[i], tat[i]);

return 0;
}
|

```

OUTPUT:-

```
praneeth@PRANEETH:~$ nano round.c
praneeth@PRANEETH:~$ gcc round.c -o round
praneeth@PRANEETH:~$ ./round
```

Enter number of processes: 3

Enter burst time:

P1: 11

P2: 22

P3: 33

Enter time quantum: 3

Process	BT	WT	TAT
P1	11	18	29
P2	22	32	54
P3	33	33	66

```
praneeth@PRANEETH:~$ |
```